

✓ Project Title:

Vulnerability Analysis and Penetration Testing | SecureBank Simulation (Final Project - IBM Cybersecurity Analyst Professional Certificate)

🔒 Project Overview:

In this capstone project, I assumed the role of an **ethical hacker** hired by *SecureBank*, a fictional financial institution, to identify and mitigate vulnerabilities in their digital infrastructure. The project was designed to simulate real-world penetration testing, threat hunting, and data encryption to enhance organizational security.

🔧 Key Tasks Completed:

🔍 Task 1: Vulnerability Identification

- Used [IBM X-Force Exchange](#) to identify a recent CVE with high severity.
- Selected a relevant vulnerability to assess its potential impact on SecureBank's systems.
- Captured and documented threat intelligence data.

🌐 Task 2: Google Dorking Investigation

- Applied **Google Dorking** techniques to further investigate the chosen CVE:
 - Queried general indexed content using: `site: CVE-2024-XXXX`
 - Narrowed focus to government resources using: `site:gov CVE-2024-XXXX`
 - Retrieved PDF-based reports using: `site:gov CVE-2024-XXXX filetype:pdf`
- Captured evidence of publicly exposed vulnerability reports.

🛡️ Task 3: Penetration Testing Plan

- Created a 6-step penetration testing strategy aligned with SecureBank's risk profile.
- Decisions were made based on:
 - Scope definition
 - Information gathering
 - Tool selection
 - Prioritization of vulnerabilities
 - Exploitation tactics
 - Executive reporting
- Justified every strategic decision with industry best practices.

Secure Data Handling Exercise (Encryption Project)

To simulate real-world post-exploitation response, I handled plaintext credentials found on a web server:

File Content:

plaintext

CopyEdit

Username: admin

Password: P@ssw0rd123

Email: admin@example.com

Phone: 123-456-7890

Steps Performed:

1. **Created the file** `userdetails` with the above credentials.
2. **Generated a 256-bit AES encryption key** and stored it in `passkey`.
3. **Encrypted the file** to `userdetails_encrypt` using the key.
4. **Displayed binary content** of the encrypted file (with caution to avoid terminal corruption).
5. **Decrypted the file** back to `userdetails_decrypt` using the same AES key.
6. **Verified the decrypted content** matched the original data.

Each step was validated with screenshots to simulate incident response documentation.

Skills Demonstrated:

- Vulnerability Intelligence Gathering (IBM X-Force)
 - Reconnaissance using Google Dorking
 - AES Encryption and Decryption (Linux + OpenSSL)
 - Penetration Testing Lifecycle Planning
 - Secure Data Handling Practices
 - Report Writing and Documentation for Technical + Executive Audiences
-

Tools Used:

- IBM X-Force Exchange
 - Google Advanced Search Operators
 - Linux Terminal (Ubuntu-based Cloud IDE)
 - OpenSSL
 - Bash Commands
-

Artifacts Available:

Screenshots for each milestone (vulnerability discovery, dorking queries, encryption steps) have been documented and submitted as part of the peer-reviewed project.

FileEditSelectionViewGoRunTerminalHelp

< >

☐

S... ?

Welcome ×

> DATABASES

> BIG DATA

> CLOUD EMBEDDABLE

> AI

> OTHER

Launch...

Cloud IDE

Version 1.7.4

theia@theia-kiranmohan75: /home/project

theia@theia-kiranmohan75: /home/project × ☐ ☐

theia@theia-kiranmohan75: /home/project\$ ls

passkey userdetails userdetails_decrypt userdetails_encrypt

theia@theia-kiranmohan75: /home/project\$ rm userdetails_encrypt

theia@theia-kiranmohan75: /home/project\$ ls

passkey userdetails userdetails_decrypt

theia@theia-kiranmohan75: /home/project\$ trash-restore

bash: trash-restore: command not found

theia@theia-kiranmohan75: /home/project\$ openssl enc -aes-256-cbc -salt -

in userdetails -out userdetails_encrypt -pass file:./passkey

ls -l

*** WARNING : deprecated key derivation used.

Using -iter or -pbkdf2 would be better.

total 16

-rw-r--r-- 1 theia users 65 Apr 16 15:28 passkey

-rw-r--r-- 1 theia users 83 Apr 16 15:24 userdetails

-rw-r--r-- 1 theia users 83 Apr 16 15:31 userdetails_decrypt

-rw-r--r-- 1 theia users 112 Apr 16 15:44 userdetails_encrypt

theia@theia-kiranmohan75: /home/project\$ cat userdetails_encrypt

Salted_N0h0.001000ZHy"}00000f0X*!}غ;0iy/0l00.00)00D730f0a~040Hl006eS0!

theia@theia-kiranmohan75: /home/project\$ openssl enc -d -aes-256-cbc -in

userdetails_encrypt -out userdetails_decrypt -pass file:./passkey

ls -l

*** WARNING : deprecated key derivation used.

Using -iter or -pbkdf2 would be better.

total 16

-rw-r--r-- 1 theia users 65 Apr 16 15:28 passkey

-rw-r--r-- 1 theia users 83 Apr 16 15:24 userdetails

-rw-r--r-- 1 theia users 83 Apr 16 15:45 userdetails_decrypt

-rw-r--r-- 1 theia users 112 Apr 16 15:44 userdetails_encrypt

theia@theia-kiranmohan75: /home/project\$ cat userdetails_decrypt

Username: admin

Password: P@ssw0rd123

Email: admin@example.com

Phone: 123-456-7890

theia@theia-kiranmohan75: /home/project\$

0 0 0

Go Live

🔔

☐

FileEditSelectionViewGoRunTerminalHelp

< > []

Welcome x

S... [?] []

> DATABASES

> BIG DATA

> CLOUD EMBEDDABLE

> AI

> OTHER

Launch...

Cloud IDE

Version 1.7.4

theia@theia-kiranmohan75: /home/project

theia@theia-kiranmohan75: /home/project x [] []

theia@theia-kiranmohan75: /home/project\$ ls

passkey userdetails userdetails_decrypt userdetails_encrypt

theia@theia-kiranmohan75: /home/project\$ rm userdetails_encrypt

theia@theia-kiranmohan75: /home/project\$ ls

passkey userdetails userdetails_decrypt

theia@theia-kiranmohan75: /home/project\$ trash-restore

bash: trash-restore: command not found

theia@theia-kiranmohan75: /home/project\$ openssl enc -aes-256-cbc -salt -

in userdetails -out userdetails_encrypt -pass file:./passkey

ls -l

*** WARNING : deprecated key derivation used.

Using -iter or -pbkdf2 would be better.

total 16

-rw-r--r-- 1 theia users 65 Apr 16 15:28 passkey

-rw-r--r-- 1 theia users 83 Apr 16 15:24 userdetails

-rw-r--r-- 1 theia users 83 Apr 16 15:31 userdetails_decrypt

-rw-r--r-- 1 theia users 112 Apr 16 15:44 userdetails_encrypt

theia@theia-kiranmohan75: /home/project\$ cat userdetails_encrypt

Salted__N0h0.00l000ZHy"}00000f0X*!} ;0iy/0]00.00)00D730f0a~040Hl006eS0!

theia@theia-kiranmohan75: /home/project\$ openssl enc -d -aes-256-cbc -in

userdetails_encrypt -out userdetails_decrypt -pass file:./passkey

ls -l

*** WARNING : deprecated key derivation used.

Using -iter or -pbkdf2 would be better.

total 16

-rw-r--r-- 1 theia users 65 Apr 16 15:28 passkey

-rw-r--r-- 1 theia users 83 Apr 16 15:24 userdetails

-rw-r--r-- 1 theia users 83 Apr 16 15:45 userdetails_decrypt

-rw-r--r-- 1 theia users 112 Apr 16 15:44 userdetails_encrypt

theia@theia-kiranmohan75: /home/project\$ cat userdetails_decrypt

Username: admin

Password: P@ssw0rd123

Email: admin@example.com

Phone: 123-456-7890

theia@theia-kiranmohan75: /home/project\$ cat

0 0 0

Go Live [] []

Cloud IDE

VS Code API Version: 1.89.1



Skills
Network

Cloud IDE is a **fully-featured** Integrated Development Environment (IDE) that you'll be using to write, run, and debug your code.

Open Workspace

On your left, you'll find the instructions for the tasks you need to complete in this lab. Follow them step-by-step to achieve the objectives of this lab. You can also click the chat

```
project /home/project
```

Onen Keyboard Shortcuts

theia@theia-kiranmohan75: /home/project x

```
theia@theia-kiranmohan75:/home/project$ home/project$
bash: home/project$: No such file or directory
theia@theia-kiranmohan75:/home/project$ nano userdetails
theia@theia-kiranmohan75:/home/project$ cat userdetails
Username: admin
Password: P@ssw0rd123
Email: admin@example.com
Phone: 123-456-7890
theia@theia-kiranmohan75:/home/project$
```


FileEditSelectionViewGoRunTerminalHelp

←→

☐

S...📄?

🔍

🔗

🔧

🔧

🔧

⚙️

> DATABASES

> BIG DATA

> CLOUD EMBEDDABLE

> OTHER

🔗 Launch...

Welcome ×

Cloud IDE

Version 1.7.4

VS Code API Version: 1.89.1



Skills

Network

What is Cloud IDE?

Cloud IDE is a **fully-featured** Integrated Development Environment (IDE) that you'll be using to write, run, and debug your code.

Getting Started

Start

New File...

Open

Open Workspace

Problems

theia@theia-kiranmohan75: ~

theia@theia-kiranmohan75: /home/project ×

☐☐☐

```
theia@theia-kiranmohan75:/home/project$ cd ..
theia@theia-kiranmohan75:/home$ cd ..
theia@theia-kiranmohan75:/home$ ls
README.md      dev            lib             media           root           sys
bin            etc            lib32           mnt             run            tmp
boot           flutter_linux_3.24.1-stable.tar.xz  lib64          opt             sbin           usr
chromedriver-linux64  home          libx32         proc            srv            var
```

```
theia@theia-kiranmohan75:/home/project$ openssl rand -hex 32>passkey
bash: passkey: Permission denied
theia@theia-kiranmohan75:/home/project$ openssl rand -hex 32 > passkey
bash: passkey: Permission denied
theia@theia-kiranmohan75:/home/project$ sudo openssl rand -hex 32 > passkey
bash: passkey: Permission denied
theia@theia-kiranmohan75:/home/project$ cd /home/project$
bash: cd: /home/project$: No such file or directory
theia@theia-kiranmohan75:/home/project$ cd /home/project
theia@theia-kiranmohan75:/home/project$ ls
userdetails
theia@theia-kiranmohan75:/home/project$ openssl rand -hex 32 |tee passkey
c0a39d7ec087173d89532e5021d45df4419d7633b8892206b08b723488862792
theia@theia-kiranmohan75:/home/project$ ls -l
total 8
-rw-r--r-- 1 theia users 65 Apr 16 15:28 passkey
-rw-r--r-- 1 theia users 83 Apr 16 15:24 userdetails
theia@theia-kiranmohan75:/home/project$ cat passkey
c0a39d7ec087173d89532e5021d45df4419d7633b8892206b08b723488862792
theia@theia-kiranmohan75:/home/project$
```

0 0 0

Tool iframe

Go Live 🔔☐